

Supplement A - Baseline Δ -Window and Dynamic Selection Analysis

This supplement supports the baseline nonlinear compatibility analysis presented in the main manuscript. It provides the numerical diagnostics, control comparisons, bootstrap checks, and figure-level evidence for the Δ -window analysis of Rössler, Lorenz, and Duffing systems.

A1. Purpose and Scope

This supplement preserves the baseline Δ -window analysis developed for Rössler, Lorenz, and Duffing systems. It assembles compatibility curves, local/global organization measures, dwell-time persistence, bootstrap localization, shuffled controls, random controls, AR(1) surrogates, and phase-randomized surrogate comparisons.

A2. Baseline Systems and Δ -Window Construction

The baseline analysis uses Rössler, Lorenz, and Duffing trajectories as representative nonlinear systems. The operational threshold Δ acts as a compatibility constraint: local transitions are considered compatible when state-distance relations fall within the Δ -window. The main diagnostic curves are $T_{\text{global}}(\Delta)$, computed from global pairwise state distances, and $T_{\text{local}}(\Delta)$, computed from temporally local neighborhoods.

A3. Local/Global Organization

The local/global organization ratio $T_{\text{local}}(\Delta)/T_{\text{global}}(\Delta)$ measures whether temporal neighborhoods are more compatible than expected from global phase-space geometry alone. Values above unity support the interpretation of local temporal organization beyond static geometry. Unless otherwise noted, dwell-time values are reported in trajectory steps, while transition counts correspond to the number of detected state-domain transitions along the analyzed trajectory.

A4. Dwell-Time Persistence and Controls

Dwell-time analysis measures how long a trajectory remains within a domain before transitioning. Longer residence durations in original trajectories than in shuffled, random, AR(1), or phase-randomized controls support the interpretation that the observed organization is dynamic rather than only geometric. Bootstrap confidence intervals for contrast estimates are available in the supplementary code archive.

A5. Bootstrap and Surrogate Behavior

Bootstrap diagnostics estimate the stability of characteristic Δ^* thresholds. Surrogate and shuffled controls test whether the method detects organization in randomized or partially constrained dynamics. The control structure is essential because it distinguishes temporal organization from static distributional artifacts.

A6. Interpretation Boundaries

The Δ -window should be interpreted as an operational compatibility threshold, not as a fundamental constant. Different systems and control pipelines may produce different Δ^* thresholds, so the supplement distinguishes global, local/global, dwell-based, and control-based Δ definitions. Phase randomization destroys long-range temporal organization, shifting Δ^* toward the scale of local fluctuations.

A7. Cross-System Baseline Summary

Table A1. Cross-system baseline summary. Dwell values are expressed in trajectory steps and transition values count detected domain transitions.

System	N	Δ^*	$T_{\text{global}}(\Delta^*)$	$T_{\text{local}}(\Delta^*)$	Local/global	Dwell contrast	Transition contrast
Rössler	2600	1.547	0.402	0.811	2.014	10.888	10.946
Lorenz	1500	1.353	0.314	0.466	1.487	4.973	4.991
Duffing	5667	1.607	0.428	0.941	2.200	16.038	16.094

A8. Original vs Control Contrast Summary

Table A2. Original-versus-control contrast summary for shuffled, random, and AR(1) variants.

System	Control	Original Δ^*	Control Δ^*	Local/global contrast	Mean dwell contrast	Transition contrast
Rössler	shuffled	1.490	1.490	2.635	11.675	11.738
Rössler	random	1.490	1.915	2.532	11.716	11.780
Rössler	AR(1) surrogate	1.490	1.963	1.089	2.899	2.911
Lorenz	shuffled	1.380	1.380	3.071	7.678	7.737
Lorenz	random	1.380	2.024	3.187	9.922	10.000
Lorenz	AR(1) surrogate	1.380	1.970	1.290	2.417	2.430
Duffing	shuffled	1.624	1.624	2.271	18.338	18.413
Duffing	random	1.624	1.443	2.340	18.628	18.704
Duffing	AR(1) surrogate	1.624	1.427	0.919	4.468	4.483

A9. Integrated Figures

The following figures summarize compatibility organization, bootstrap localization, dwell-time persistence, and control contrasts across the analyzed nonlinear systems.

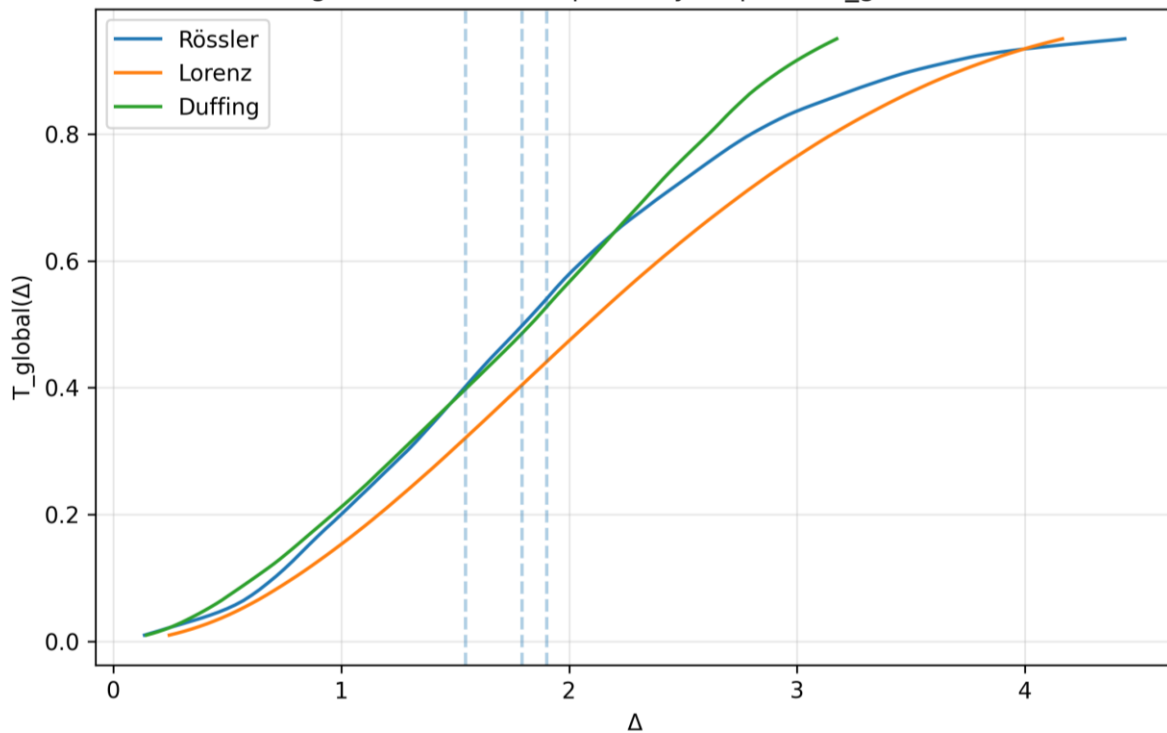


Figure A1. Mean compatibility response $T(\Delta)$.

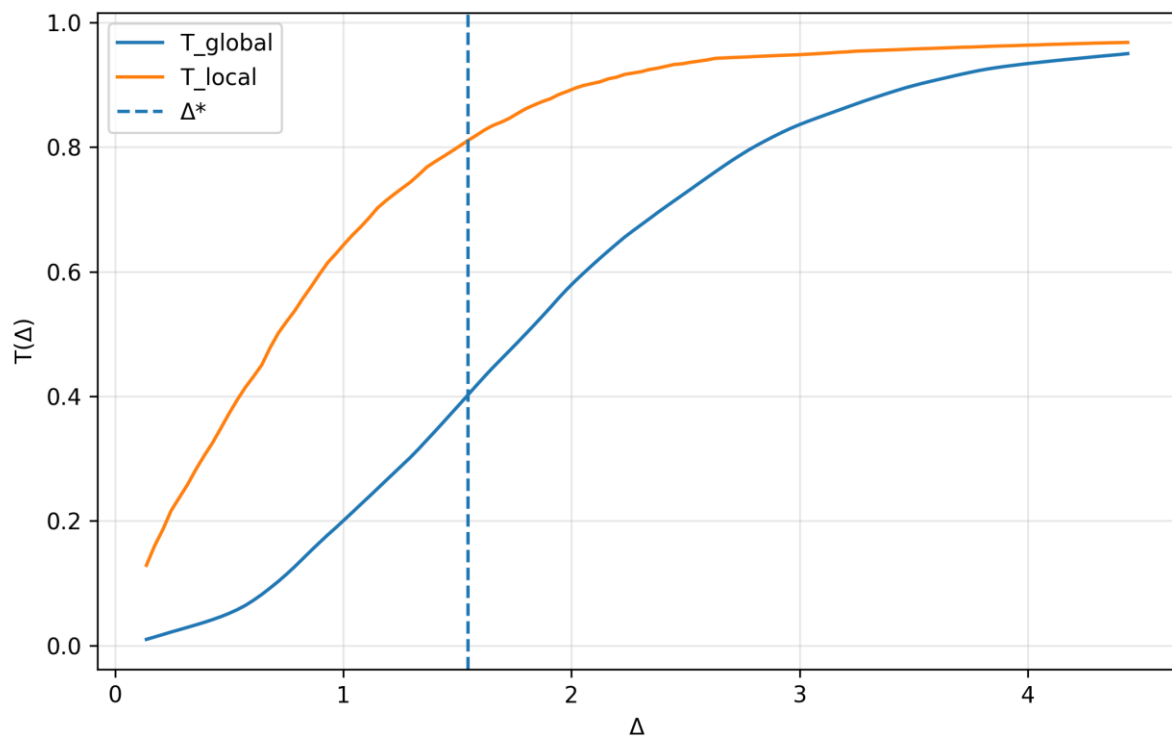


Figure A2a. Rössler local and global compatibility curves.

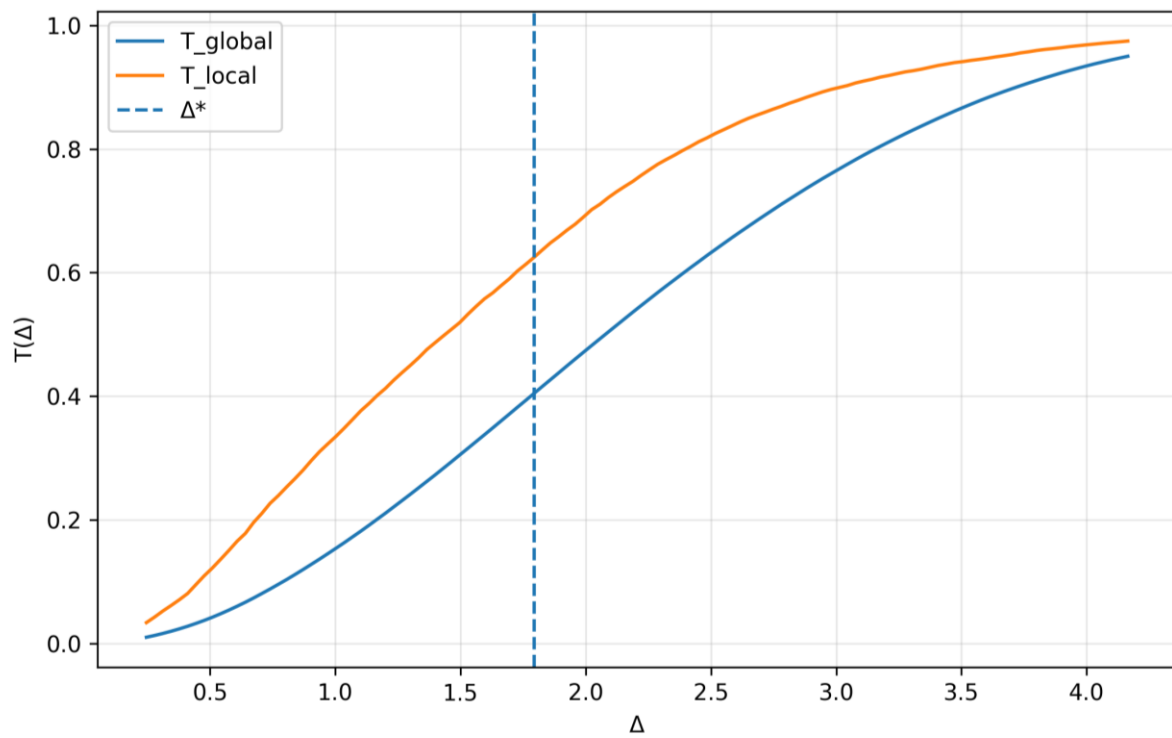


Figure A2b. Lorenz local and global compatibility curves.

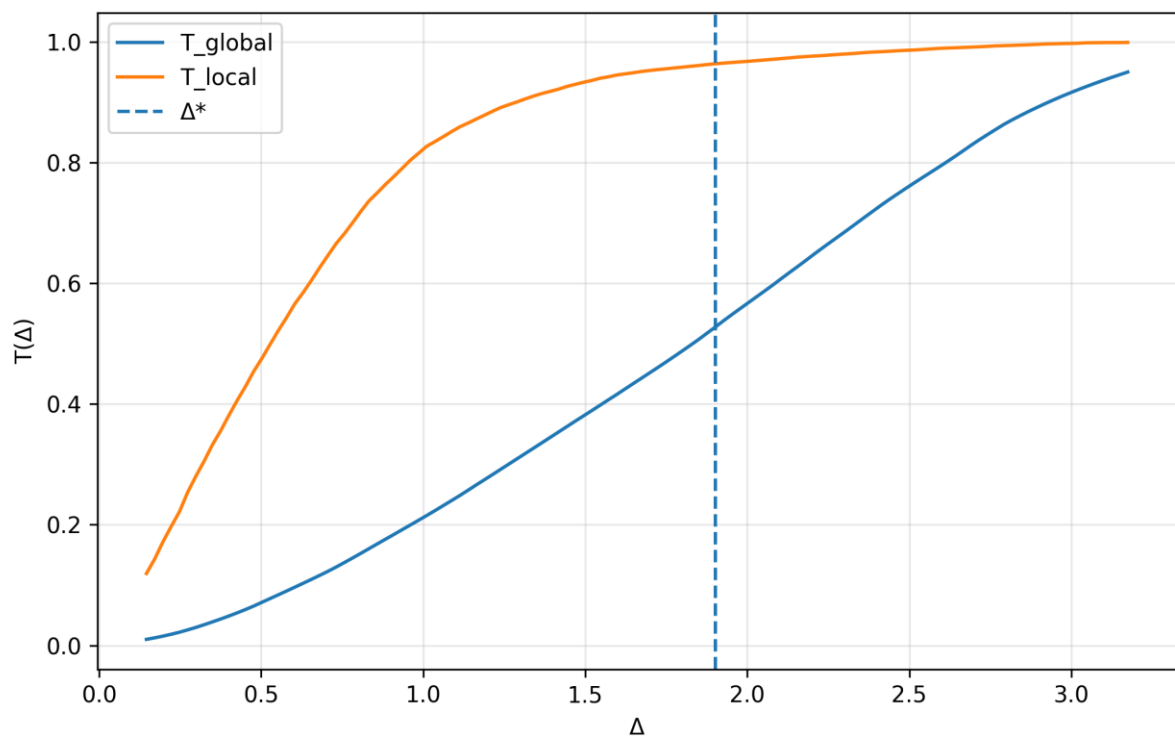


Figure A2c. Duffing local and global compatibility curves.

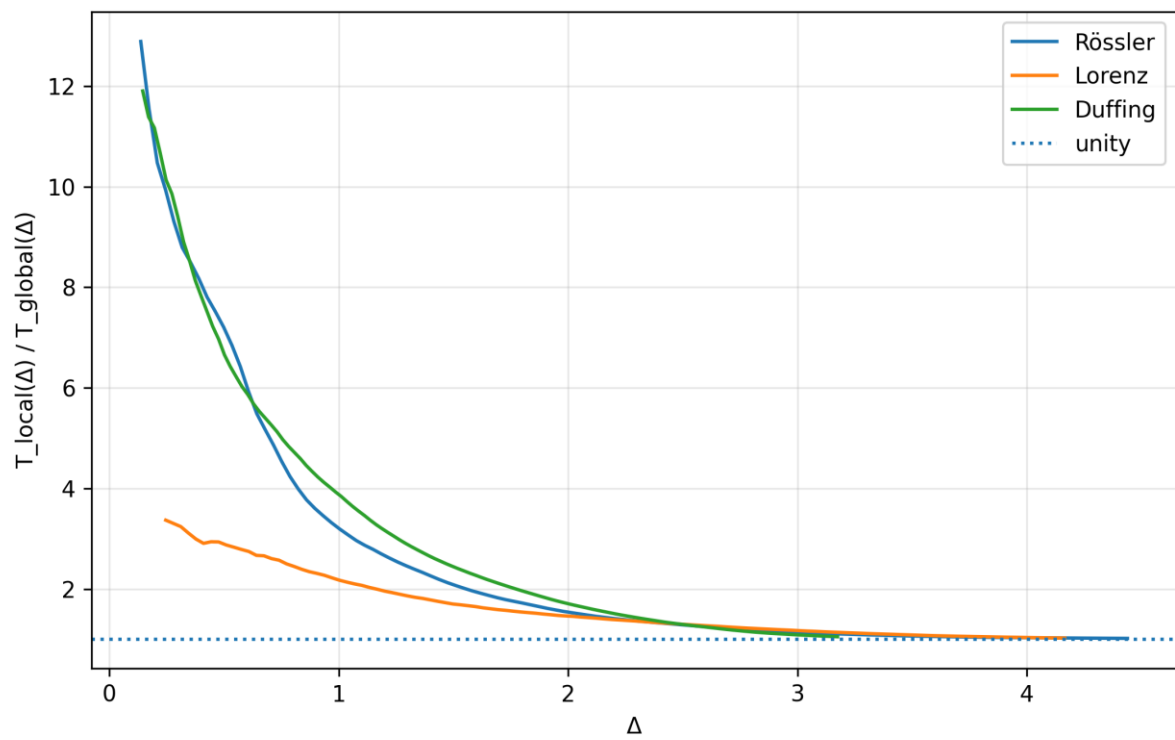


Figure A3. Local/global organization index across systems.

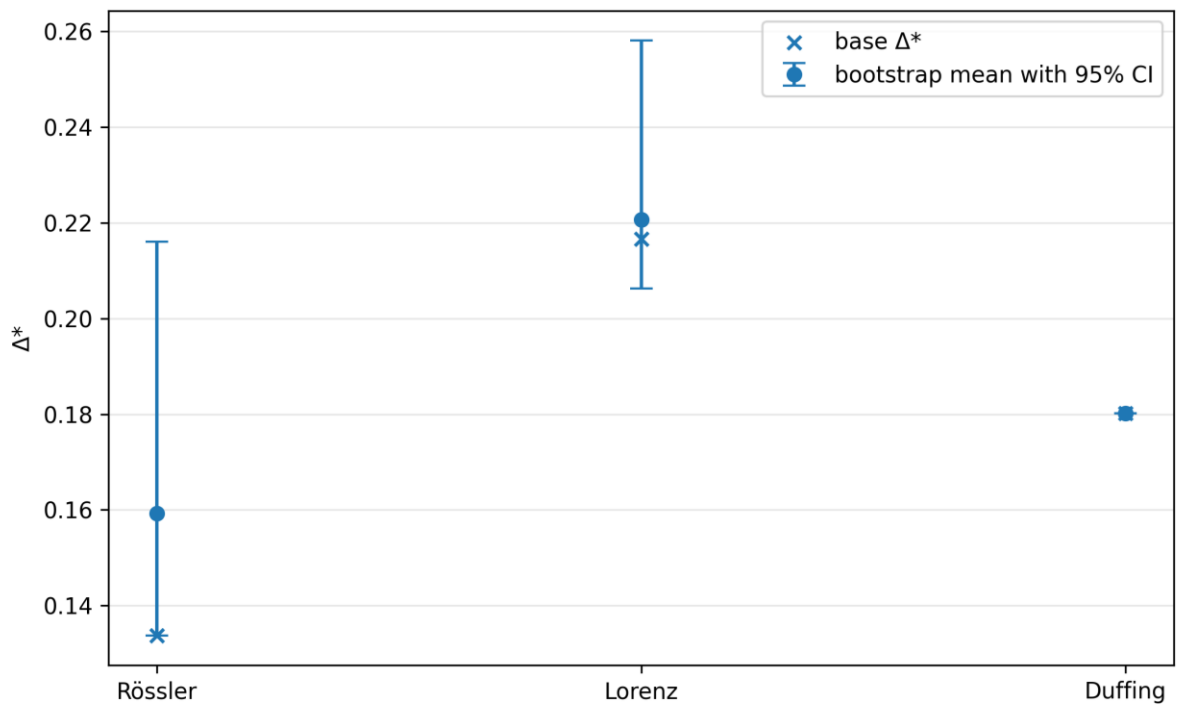


Figure A4. Bootstrap localization summary.

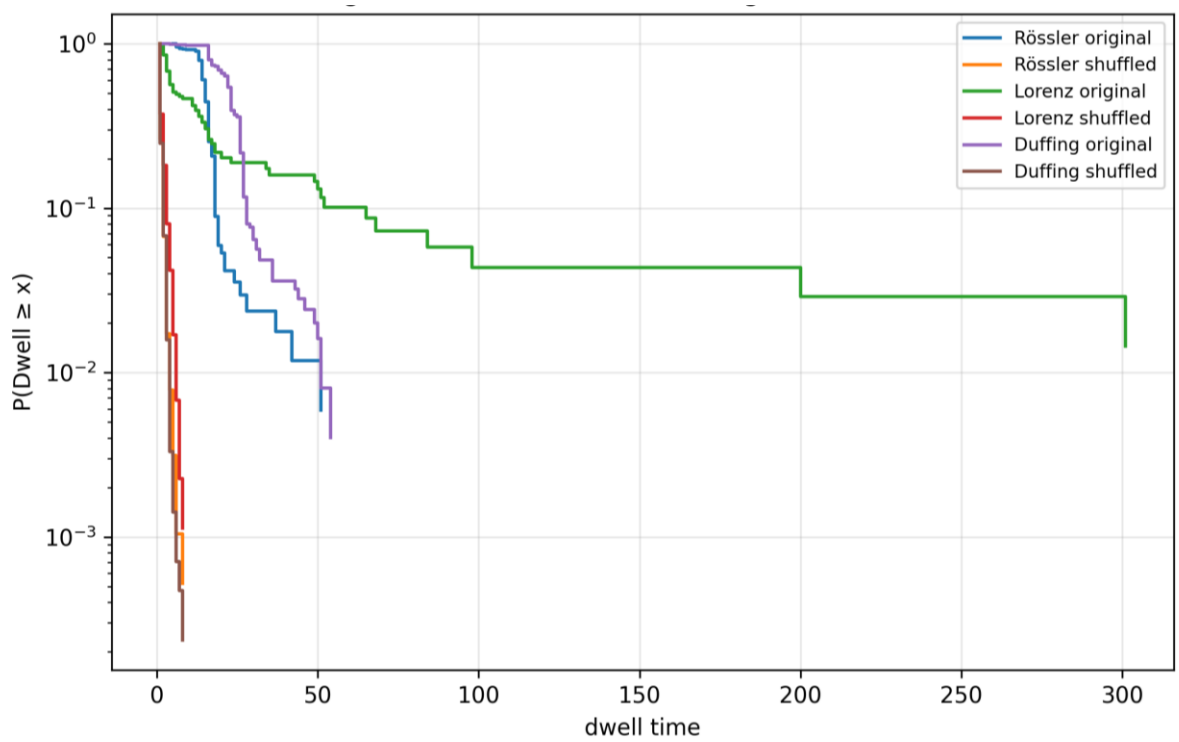


Figure A5a. Dwell-time CCDF: original vs shuffled.

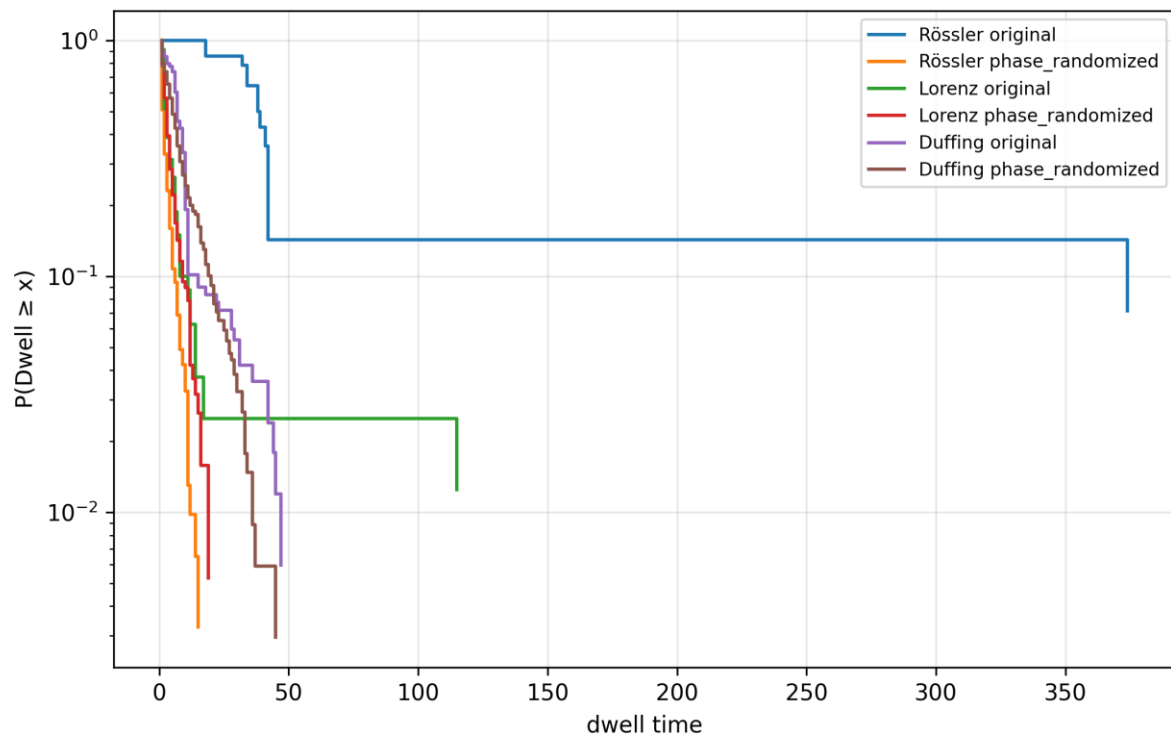


Figure A5b. Dwell-time CCDF: original vs phase-randomized.

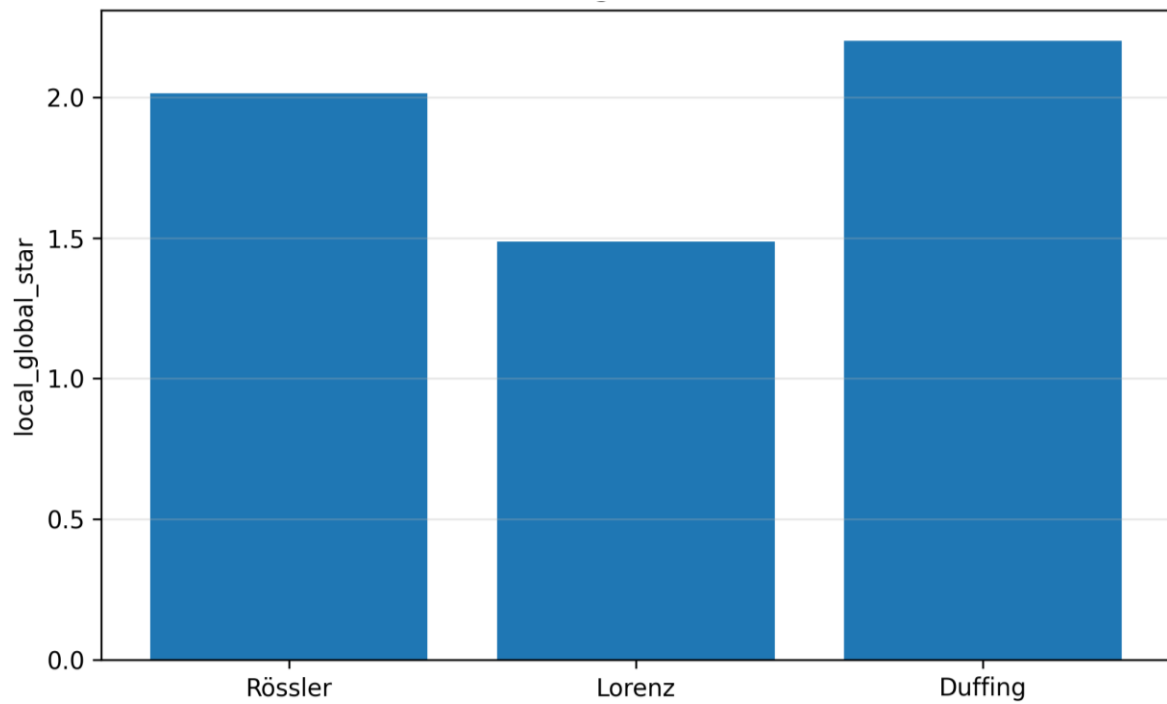


Figure A6. Local/global organization ratios at Δ^* .

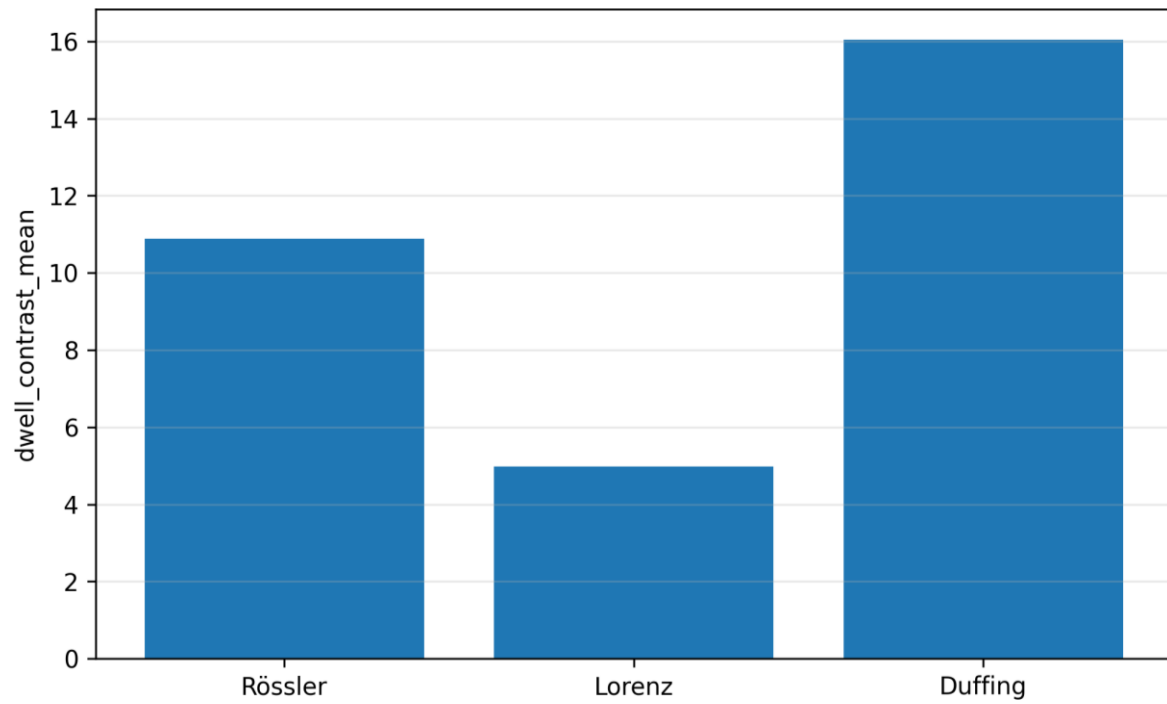


Figure A7. Dwell-time contrast summary across systems.

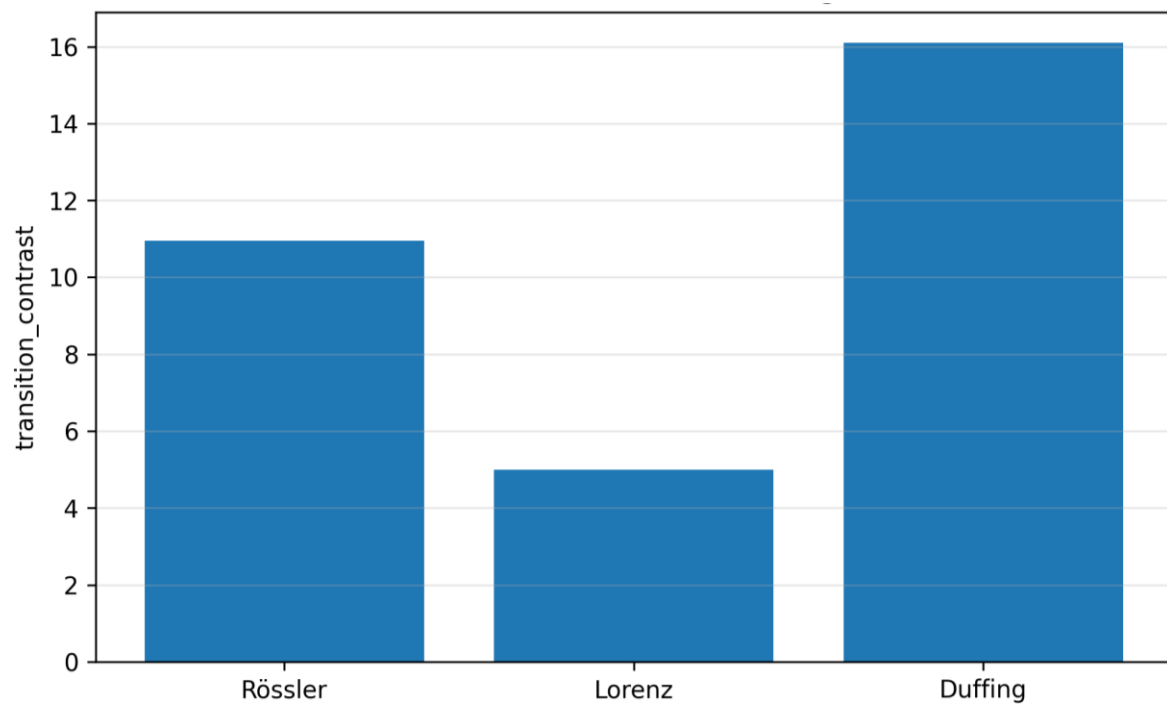


Figure A8. Transition contrast summary across systems.

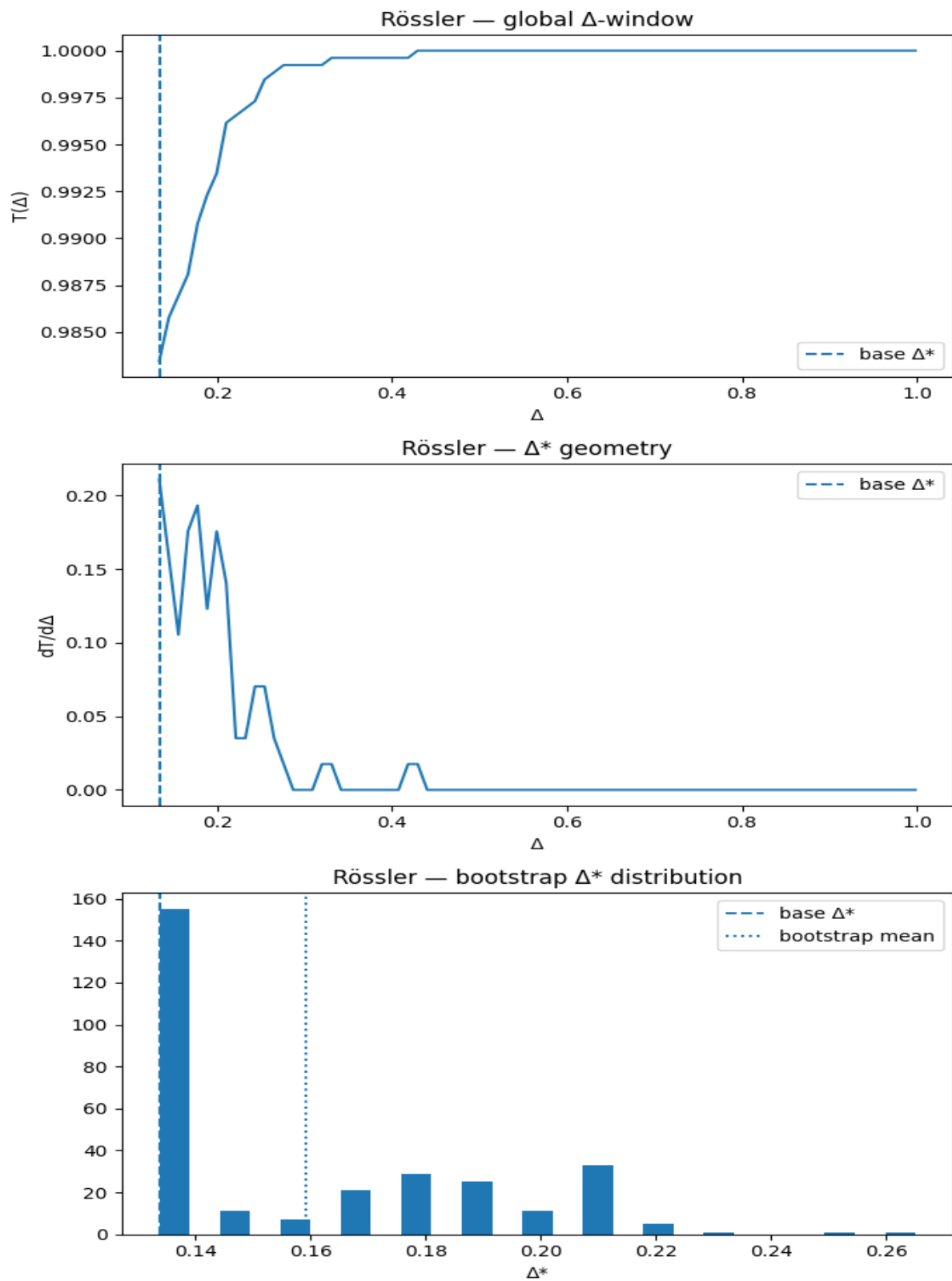


Figure A9. Bootstrap Δ^* distribution panel from the baseline diagnostic run.

A10. Current Analysis Scope and Extensions

This supplement focuses on metric-level diagnostics, compatibility structure, and control comparisons across nonlinear systems. Additional attractor-domain visualizations, phase-sector analyses, radial-bin studies, and higher-dimensional geometric representations may be included in future extensions of the analysis pipeline.

A11. Extended Tables

The extended tables provide the detailed numerical basis for the baseline results. Wide tables are placed in compact format to preserve readability.

Table A3. Core nonlinear-system metrics and compatibility organization, max $dT/d\Delta$ denotes the maximum slope of $T(\Delta)$, interpreted as the sharpness of the compatibility transition.

System	N	Δ^*	max $dT/d\Delta$	$T_{\text{global}}(\Delta^*)$	$T_{\text{local}}(\Delta^*)$	Local/global	Max local/global	Δ at max local/global
Rössler	2600	1.547	0.410	0.402	0.811	2.014	12.888	0.137
Lorenz	1500	1.353	0.357	0.314	0.466	1.487	2.341	0.285
Duffing	5667	1.607	0.395	0.428	0.941	2.200	16.717	0.182

Table A4. Dwell-time and transition contrasts for the baseline systems.

System	Mean dwell (steps)	Shuffle dwell (steps)	Dwell contrast	Transitions	Shuffle transitions	Transition contrast
Rössler	15.385	1.413	10.888	168	1839	10.946
Lorenz	6.818	1.371	4.973	219	1093	4.991
Duffing	21.305	1.328	16.038	265	4265	16.094

Table A5. Control variants for original, shuffled, random, and AR(1) surrogate trajectories.

System	Variant	Δ^*	$T_{\text{global}}(\Delta^*)$	$T_{\text{local}}(\Delta^*)$	Local/global	Mean dwell (steps)	Transitions
Rössler	original	1.490	0.379	0.979	2.581	15.385	168
Rössler	shuffled	1.490	0.379	0.372	0.980	1.318	1972
Rössler	random	1.915	0.391	0.399	1.019	1.313	1979
Rössler	AR(1) surrogate	1.963	0.422	1.000	2.370	5.306	489
Lorenz	original	1.380	0.323	0.985	3.049	13.043	114
Lorenz	shuffled	1.380	0.323	0.321	0.993	1.699	882
Lorenz	random	2.024	0.437	0.418	0.957	1.315	1140
Lorenz	AR(1) surrogate	1.970	0.423	1.000	2.363	5.396	277
Duffing	original	1.624	0.434	1.000	2.303	24.532	230
Duffing	shuffled	1.624	0.434	0.440	1.014	1.338	4235
Duffing	random	1.443	0.407	0.401	0.984	1.317	4302
Duffing	AR(1) surrogate	1.427	0.399	1.000	2.506	5.491	1031

Table A6. Phase-randomized compatibility contrasts.

System	Original Δ^*	Phase-randomized Δ^*	Δ^* difference	Original local/global	Phase-randomized local/global	Local/global contrast	Original mean dwell (steps)	Phase-randomized mean dwell (steps)	Mean dwell contrast	Original transitions	Phase-randomized transitions	Transition contrast
Rössler	0.090	0.203	-0.113	1.009	0.994	1.015	56.714	2.616	21.683	1805	1796	0.995
Lorenz	0.230	0.328	-0.098	1.002	0.994	1.007	5.429	2.508	2.164	1081	1035	0.957
Duffing	0.067	0.080	-0.013	0.998	0.998	1.001	11.551	7.898	1.463	4095	3494	0.853

Table A7. Bootstrap statistics and Δ^* localization from the baseline diagnostic run. Bootstrap N denotes the number of resamples. The coarse-grid Duffing bootstrap locking effect is revisited in the high-resolution reanalysis presented in Table A8.

System	Base Δ^*	Base max $dT/d\Delta$	Bootstrap mean Δ^*	Bootstrap median Δ^*	Bootstrap SD Δ^*	CI95 low	CI95 high	Bootstrap N
Rössler	0.134	0.211	0.159	0.134	0.031	0.134	0.216	300
Lorenz	0.217	0.354	0.221	0.217	0.018	0.206	0.258	300
Duffing	0.180	0	0.180	0.180	0	0.180	0.180	300

Table A8. High-resolution Duffing bootstrap reanalysis. This reanalysis used 500 block-bootstrap realizations and a dense Δ -grid.

System	Definition	Original Δ^*	Bootstrap mean Δ^*	Bootstrap median Δ^*	Bootstrap SD Δ^*	CI95 low	CI95 high	Bootstrap N
Duffing	global Δ^*	2.010	1.779	1.823	0.268	1.378	2.333	500
Duffing	local Δ^*	0.085	0.084	0.085	0.021	0.055	0.119	500

A12. Global vs Local Δ^* Definitions

The supplement distinguishes between two operational forms of Δ^* . Global Δ^* corresponds to large-scale compatibility reorganization across the attractor geometry, whereas local Δ^* describes short-range temporal compatibility between neighboring trajectory states. These quantities are computed on different effective scales and therefore are not expected to be numerically identical.

Additional Duffing bootstrap reanalysis confirmed that the previously observed zero-variance bootstrap result originated from insufficient Δ -grid resolution and discretization locking. Using a denser Δ -grid together with block-bootstrap resampling produced stable but nonzero bootstrap variance: global Δ^* mean about 1.779 with SD about 0.268; local Δ^* mean about 0.084 with SD about 0.021. Table A1 reports the original global $T(\Delta)$ scan estimate, whereas Table A8 reports the refined high-resolution bootstrap reanalysis.

The Duffing system therefore exhibits distinct global and local Δ^* regimes, suggesting partial separation between attractor-scale reorganization and local temporal compatibility dynamics.

A13. Interpretation Summary

Across the analyzed nonlinear systems, the original trajectories consistently exhibit stronger local/global compatibility organization, longer dwell persistence, and reduced transition fragmentation relative to shuffled and randomized controls. AR(1) surrogates partially preserve short-range temporal correlations but do not reproduce the full compatibility structure observed in the original systems. These results support the interpretation of the Δ -window as a robust compatibility regime associated with dynamic organization rather than static geometric coincidence.

Publication formatting note: the supplementary material was standardized for publication formatting, including figure integration, table normalization, terminology harmonization, bootstrap clarification, and removal of exploratory workflow artifacts originating from intermediate notebook-based analyses.